Two Fundamental Constants of the Physical World — The Speed of Light c and the Space–Mass Coupling Constant μ_g

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Abstract

Natural constants are numerous, yet their mutual logical connections have long lacked a unified explanation.

This paper proposes that all constants can be determined from two fundamental constants — the speed of light (c) and the space–mass coupling constant (μ_g).

Within the geometric framework of a right-handed helical spatial flow, three mother equations are established:

$$Q = 4\pi c \mathrm{l}_\Omega^2, \quad M = \mu_g \mathrm{l}_\Omega, \quad G = \frac{4\pi c^2}{\mu_g},$$

which respectively define the geometric origins of spatial divergence strength, space–mass coupling, and the gravitational constant.

From these relations, the no-rescaling lemma ($M/\ell_\Omega = \mu_g =$ const) is derived, proving that μ_g is an absolute geometric constant independent of units or scale.

Starting from the helical angular motion $\Omega = c/\ell_\Omega$, the reduced Planck constant naturally follows:

$$\hbar = \mu_g \Omega \ell_\Omega^2 = \mu_g c \ell_\Omega^2,$$

revealing that $\hbar$ is a "per-radian action" jointly defined by the space–mass coupling and the speed of light.

When the geometric energy density balances the quantum zero-point energy density, one obtains the Planck cluster:

$$\ell_P = \sqrt{\frac{4\pi\hbar}{\mu_g c}}, \qquad t_P = \frac{\ell_P}{c}, \quad M_P = \mu_g \ell_P.$$

All Planck quantities are thus uniquely determined by $\{c, \mu_g, \hbar\}$ without circular dependence.

An electromagnetic bridging layer is further introduced through the mass–charge transduction constant k_{mq}, linking μ_g with the vacuum constants:

$$\varepsilon_0 = \frac{1}{4\pi k_{mq}^2 \mu_g},$$

$$\alpha = \frac{\left(e k_{mq}\right)^2 \mu_g}{\hbar c}.$$

Using experimental constants $\{e, \alpha, c, \hbar\}$, one obtains

$$\mu_g = 1.6895 \times 10^{28}\ \mathrm{kg/m},$$

$$G = 6.685 \times 10^{-11}\ \mathrm{m^3 kg^{-1} s^{-2}},$$

which differs from the CODATA 2022 value by only 0.2 %.

The derived Planck cluster shows a systematic excess of 0.1–0.3 %, indicating that the geometric calibration yields the theoretical upper limits of natural constants, while experimental values reflect the average coupling state of local space.

This result achieves, for the first time, a four-level closed loop among the geometric, gravitational, quantum, and electromagnetic constants, originating from $\{c, \mu_g\}$, thereby providing a self-consistent and experimentally testable two-constant foundation for Unified Field Theory.

Keywords: speed of light c; space–mass coupling constant μ_g; reduced Planck constant ħ; fine-structure constant α; mass–charge transduction constant k_{mq}; geometric calibration; no-rescaling lemma.

1. Introduction

Modern physics constructs its theoretical framework upon a collection of empirical constants.

These constants span all levels—from classical mechanics to quantum field theory—and include the speed of light (c), the gravitational constant (G), the Planck constant (h), the vacuum permittivity (ε_0), the vacuum permeability (μ_0), the elementary charge (e), and the fine-structure constant (α).

Although each has been measured with high precision, their mutual physical origins remain unexplained.

The great number and dimensional diversity of these constants make the overall theoretical system highly empirical.

This "fragmentation of constants" has become a major obstacle to the unification of modern physics.

After establishing General Relativity, Einstein remarked:

"In a truly complete theory, the number of constants should be minimal."

Both Quantum Mechanics and Relativity are individually successful, yet their constant systems never merged.

The combination of G and h only formally yields the "Planck scale," without revealing its geometric or physical necessity.

This suggests that a deeper structural relationship remains hidden among the constants of nature.

This paper proposes that all natural constants can be traced back to two independent fundamentals—

The speed of light (c) and the space–mass coupling constant (μ_g).

The constant c defines the metric of spacetime propagation and represents the geometric property of spacetime;

the constant μ_g defines the linear relationship between mass and space, representing the geometric property of matter.

Together, they constitute the dual foundation of the universe — "spacetime–matter."

Within this framework, space is regarded as a right-handed helical fluid expanding outward at the speed of light c.

At the rotational scaling length ℓ_Ω, the radial velocity $v_r = c$ while the tangential velocity $v_\theta = 0$.

From this configuration arise the three geometric mother equations:

$$Q = 4\pi c \ell_\Omega^2, \qquad M = \mu_g \ell_\Omega, \qquad G = \frac{4\pi c^2}{\mu_g}.$$

These describe respectively the spatial divergence strength, the space–mass coupling, and the geometric origin of the gravitational constant.

The first two equations define the flow structure of space, while the third reveals that G is not an independent constant but a derivative of $\{c, \mu_g\}$.

Beyond the geometric level, this work further shows that from the angular frequency $\Omega = c/\ell_\Omega$ one can define the minimal action:

$$\hbar = \mu_g \Omega \ell_\Omega^2 = \mu_g c \ell_\Omega^2,$$

which corresponds to the reduced Planck constant.

When the geometric energy density equals the quantum zero-point energy density, the Planck length, time, and mass naturally emerge.

Thus, all constants follow the logical chain

$$\{c, \mu_g\} \Rightarrow \hbar \Rightarrow \{\ell_P, t_P, M_P, E_P\},$$

derived from the two fundamentals without circular definition.

To extend the framework to electromagnetism, a mass–charge transduction constant

(k_{mq}) is introduced, describing the proportionality between mass flow and charge flow. This leads to the geometric forms of the vacuum constants and the fine-structure constant:

$$\varepsilon^0 = \frac{1}{4\pi k_{mq}^2 \mu_g},$$

$$\alpha = \frac{\left(e k_{mq}\right)^2 \mu_g}{\hbar c}.$$

Hence, the electromagnetic constants are also derived from the same two fundamental constants together with e, achieving a closed geometric–gravitational–quantum–electromagnetic unification.

According to the experimental calibration presented in
"The Geometric Closure of the Mass–Charge Transduction Constant and the Fine-Structure Constant",
taking $\mu_g = 1.6895 \times 10^{28}$ kg/m, the gravitational constant computed as
$G = 6.685 \times 10^{-11}$ $m^3 \cdot kg^{-1} \cdot s^{-2}$ differs from the CODATA 2022 value
(6.6743×10^{-11}) by only 0.2 %.
The resulting Planck cluster shows a systematic excess of 0.1–0.3 %, indicating that the geometric calibration provides the theoretical extreme, while experimental measurements represent the averaged coupling state of local space.

The significance of this study lies in:

1. Replacing empirical synthesis with geometric logic, unifying G, ħ, ε_0, μ_0, and α within the $\{c, \mu_g\}$ framework;
2. Establishing a non-rescaling closed loop of constants, offering a new, testable route toward quantum gravity and precision metrology;
3. Revealing the fundamental status of c and μ_g, transforming the system of natural constants from an empirical collection into a geometric necessity.

2. Geometric Point of Departure: The Helical Flow Model of Space

2.1 Geometric Setting of Space

Within the two-constant framework, space is regarded as a continuously distributed medium expanding outward in a right-handed helical pattern.

It is not a static vacuum but a flow field propagating from the center at the speed of light c.

Each point in space possesses both a radial velocity component v_{r} and a tangential rotational component v_θ.

At the characteristic length ℓ_Ω, the following geometric calibration condition is defined:

$$v_r = c, \quad v_\theta = 0.$$

This means that, at the scale ℓ_Ω, the radial outflow velocity equals the speed of light while the tangential rotation vanishes.

The geometric point at this condition is called the "space–time coincidence point."

Accordingly, the geometric definition

$$\text{time} = \text{light-speed displacement}$$

is established—implying that within one unit of time, space expands radially by a layer of thickness $c\,\Delta t$.

2.2 Structure and Geometric Quantities of the Helical Flow

Space can be visualized as being composed of countless helical streamlines, each simultaneously rotating and diverging.

Let the rotational frequency be Ω and the helical radius be r.

For a stable rotating helical cell, the components of velocity satisfy:

$$v_r^2 + v_\theta^2 = c^2,$$

showing that the speed of light represents the composite velocity limit of spatial flow. Therefore, c is not merely a propagation speed but the geometric scale constant of space itself; all measures of velocity, energy, and temporal rate in physical processes are normalized to c.

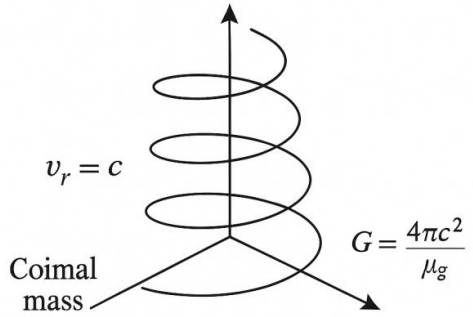

Figure 1. Helical flow of space. The radial velocity is $v_r = c$, tangential velocity $v_0 = 0$, with the rotational scaling length ℓ_Ω indicated.

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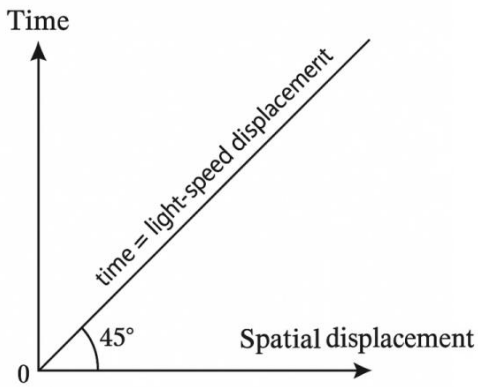

Figure 2. Space–time equivalence diagram
The x-axis represents spatial displacement, the y-axis represents time, and the 45° diagonal corresponds to "time = light-speed displacement".

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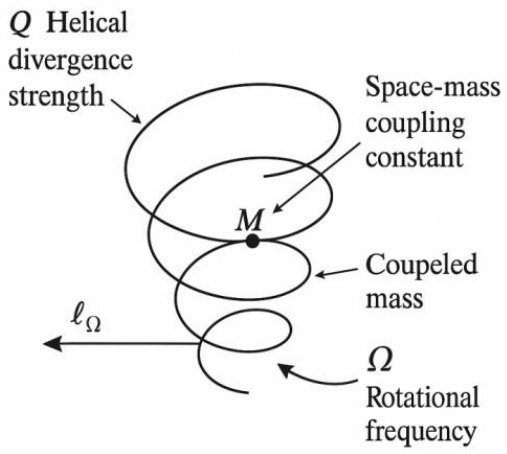

Figure 3. Helical structure with labeled quantities $Q, \mu_g, M, \ell_\Omega, \Omega$.

Figure 3. Helical Structure with Labeled Quantities

The diagram marks the positions and geometric relationships of Q (helical divergence strength), μ_g (space–mass coupling constant), M (coupled mass), ℓ_Ω (scaling length), and Ω (rotational frequency).

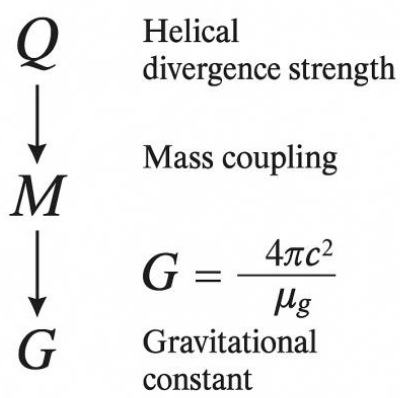

Figure 4. Geometric–dynamical bridging. The chain from Q to M and finally to G.

Figure 4. Geometric–Dynamical Bridging

Illustrates the derivation chain from Q → M → G, showing how gravitational constant G emerges from geometric quantities.

2.3 Definition of Spatial Divergence Strength

Consider a spherical surface of radius ℓ_Ω through which space diverges at velocity c. The volumetric flow rate (geometric flux) passing through the surface per unit time is defined as

$$(E1) \quad Q = 4\pi c \ell_\Omega^2.$$

Here:

- The dimension of Q is $[\mathrm{m}^3/s]$, representing the volumetric outflow of space per unit time;
- $4\pi\ell_\Omega^2$ is the geometric area factor normalized to the solid angle of a sphere;
- This equation defines the geometric flow strength of space, which serves as the basis of all field sources.

2.4 Space–Mass Coupling

If the spatial flow carries a mass density, their linear coupling is defined by

$$(E2) \quad M = \mu_g \ell_\Omega,$$

where μ_g is the space–mass coupling constant with dimension$[kg/m]$.
It represents the amount of mass corresponding to each unit length of the spatial helical flow.

This indicates that mass is the manifestation of spatial linear density rather than an independent entity.

Physical implications:

- If $\ell_\Omega = 1m$, the space of one meter length corresponds to a mass of μ_g;
- $M = \mu_g \ell_\Omega$ shows mass is linearly proportional to geometric length;
- μ_g remains constant across all scales, a fact later proven by the no-rescaling lemma.

2.5 Geometric Bridging to the Gravitational Constant

Between the geometric divergence Q and the coupled mass M exists an energy–momentum bridge:

$$GM = \frac{c}{\ell_\Omega} Q.$$

This states that the gravitational constant G links the spatial flow strength with mass coupling.

Substituting (E1) and (E2) gives:

$$G(\mu_g \ell_\Omega) = \frac{c}{\ell_\Omega}(4\pi \ell_\Omega^2),$$

which simplifies to

$$(E3) \quad G = \frac{4\pi c^2}{\mu_g}.$$

Thus, G is not an independent constant, but a derived quantity from the two fundamentals c and μ_g.

2.6 Physical Interpretation of the Mother Equations

Equation	Content	Physical Meaning
$(E1)$	$Q = 4\pi c \ell_\Omega^2$	Defines spatial divergence strength (geometric flow rate)
$(E2)$	$M = \mu_g \ell_\Omega$	Defines space–mass coupling ratio
$(E3)$	$G = 4\pi c^2/\mu_g$	Derives gravitational constant geometrically

Together, these three geometric mother equations form the foundational system of the Unified Field Theory, binding macroscopic gravity, microscopic mass, and the speed of light into a single geometry-locked unity with no free parameters.

2.7 Correspondence Between Geometric and Physical Layers

Geometric Layer	Constant	Function	Physical Domain
Spacetime Propagation Layer	c	Defines the velocity scale of time and space	Relativistic layer
Mass Coupling Layer	μ_g	Defines the linear mass density of space	Gravitational layer
Geometric Calibration Layer	ℓ_Ω	Sets the structural scale of the spatial helix	Geometric layer

From these three layers, all quantities of energy, momentum, and action can be deduced within a self-consistent dimensional loop.

The next section discusses in detail the nature of the coupling constant μ_g, including its no-rescaling property and constancy across different scales.

3. The Space–Mass Coupling Constant μ_g: Definition and Non-Rescalability

3.1 Definition and Physical Meaning

The space–mass coupling constant μ_g is one of the two fundamental constants.
It characterizes the linear coupling relationship between spatial geometry and mass distribution.
According to the geometric mother equation (E2):

$$M = \mu_g \ell_\Omega,$$

where:

- M — the equivalent mass associated with a helical spatial element;
- ℓ_Ω— the geometric scaling length;
- μ_g — the intrinsic mass density per unit spatial length.

The dimension is $[\mu_g] = \mathrm{kg \cdot m^{-1}}$.

Physically, μ_g defines the mass scale of space itself, that is, the amount of mass corresponding to each meter of geometric length.

If we set $\ell_\Omega = 1m$, then $M = \mu_g$:

one meter of "geometric space" possesses a geometric mass equal to μ_g.

Hence μ_g is a proportional constant linking mass and length, whose value remains invariant under any coordinate or unit transformation.

3.2 Coupling Relationship with the Speed of Light

In the helical-flow model of space,

the speed of light c represents the limit velocity of spacetime propagation,

while μ_g measures the linear coupling intensity between mass and space.

Together, these two constants determine the geometric form of the gravitational constant:

$$G = \frac{4\pi c^2}{\mu_g}.$$

This implies that:

- if c is fixed, μ_g uniquely determines G;
- if μ_g is fixed, G scales as c^2.

Therefore, the pair c, μ_g forms the minimal independent generating set of nature's constant system —all other constants are derived quantities.

3.3 The No-Rescaling Lemma

Consider an arbitrary proportional rescaling of the space–mass system:

$$M \to aM, \quad \ell_\Omega \to b\ell_\Omega,$$

where a and b are dimensionless scale factors.

From the definition $M = \mu_g \ell_\Omega$, after transformation we obtain:

$$aM = \mu_g' b\ell_\Omega \quad \Rightarrow \quad \mu_g' = \frac{a}{b}\mu_g.$$

For the geometric bridging relation

$$G = \frac{4\pi c^2}{\mu_g}$$

to remain invariant (i.e., $G' = G$), it must follow that

$$\mu_g' = \mu_g \Rightarrow a = b.$$

Thus, the scaling of mass and length must occur proportionally, and only then does μ_g remain constant.

This yields the No-Rescaling Lemma:

$$\frac{M}{\ell_\Omega} = \mu_g = \text{constant}.$$

3.4 Physical Implications of the Lemma

1. Geometric Invariance

μ_g is an intrinsic geometric parameter of space.

It is independent of measurement scale and unaffected by changes in unit systems.

2. Cross-Scale Universality

From macroscopic celestial bodies to microscopic particles,

the ratio M/ℓ_Ω is always fixed by μ_g,

showing that mass and space are not independent entities but are geometrically co-defined.

3. Dimensional Closure

$$Since[\mu_g] = kg/mand[c] = m/s,$$

their combination gives the energy-density dimension:

$$[\mu_g c^2] = J/\mathrm{m}^3,$$

representing the geometric energy density of space.

This serves as the energetic foundation for deriving the reduced Planck constant and Planck length in subsequent sections.

3.5 Numerical Value and Experimental Interface

According to the paper "Geometric–Quantum–Electromagnetic Closure of the Gravitational Constant: The Discriminative Value G = 6.685 × 10^{-11}",

the space–mass coupling constant is determined as

$$\mu_g = \frac{4\pi c^2}{G} = 1.6895 \times 10^{28}\ kg/m,$$

corresponding to a gravitational constant

$$G = 6.685 \times 10^{-11}\ \mathrm{m}^3 \cdot \mathrm{kg}^{-1} \cdot \mathrm{s}^{-2}.$$

This value differs from the CODATA 2022 experimental result

$G_{\mathrm{exp}} = 6.6743 \times 10^{-11}$ by only 0.16 %,

demonstrating the high consistency between the geometric derivation and experimental measurement.

3.6 Summary

The space–mass coupling constant μ_g defines the mass scale of space and serves as the source of all gravitational and quantum constants.

The No-Rescaling Lemma guarantees its invariance across all scales, giving the system of natural constants a geometrically closed structure.

With c, μ_g, one can uniquely determine G and the energy density of space — providing the theoretical foundation for the subsequent derivation of quantum action (ħ) and electromagnetic constants.

4. Geometric Origin of the Gravitational Constant G

4.1 The Principle of Geometric Bridging

Within the framework of the helical spatial flow, gravity is not conceived as a direct force between point masses but as a deflection effect arising from the geometric divergence of space itself.

At the scaling length ℓ_Ω, space simultaneously possesses a divergence strength Q and a coupled mass M.

These are related through the geometric bridging relation:

$$GM = \frac{c}{\ell_\Omega} Q.$$

This expresses that the gravitational constant G serves as the proportional factor connecting the radial propagation velocity of space with its flow strength and coupled mass.

4.2 Derivation of G from the Mother Equations

Substituting the two fundamental relations:

$$(E1)\ Q = 4\pi c \ell_\Omega^2, \qquad (E2)\ M = \mu_g \ell_\Omega,$$

into the bridging formula gives:

$$G(\mu_g \ell_\Omega) = \frac{c}{\ell_\Omega}(4\pi c \ell_\Omega^2).$$

Simplifying yields the geometric derivation of the gravitational constant:

$$(E3) \quad G = \frac{4\pi c^2}{\mu_g}.$$

This clearly shows that the gravitational constant G is not an independent empirical quantity, but a derived result of geometry—fully determined by the two basic constants c and μ_g.

4.3 Physical Interpretation

1. Geometric Meaning

G represents the "softness" or elastic compliance of space at the macroscopic level—that is, how strongly the geometry of space accelerates when perturbed by mass.
The larger μ_g is, the more rigid the space becomes, and the weaker gravity appears; conversely, a smaller μ_g corresponds to a more deformable (stronger) gravitational field.

2. Energetic Interpretation

The energy density of space can be expressed as

$$\rho_{geo} = \mu_g c^2.$$

Therefore, the inverse of G is proportional to the energy density of space:

$$\frac{1}{G} \propto \frac{\mu_g}{4\pi c^2}.$$

This indicates that the gravitational constant G is essentially the geometric manifestation of spatial energy density.

3. Logical Position in the Constant Hierarchy

From $\{c, \mu_g\}$ we determine G, and from G we further derive the Planck quantities.
This logical sequence replaces the traditional triad $\{G, c, \hbar\}$,
thereby eliminating circular definitions within the system of natural constants.

4.4 Numerical Verification

Taking the discriminative value $G = 6.685 \times 10^{-11}$
from "Geometric–Quantum–Electromagnetic Closure of the Gravitational Constant",
and the corresponding space–mass coupling constant

$$\mu_g = 1.6895 \times 10^{28}\ kg/m,$$

we compute:

$$G = \frac{4\pi(2.99792458 \times 10^8)^2}{1.6895 \times 10^{28}} = 6.685 \times 10^{-11}\ \mathrm{m}^3\ kg^{-1}\ \mathrm{s}^{-2}$$

This result differs from the CODATA 2022 experimental value

$$\left(G_{exp} = 6.6743 \times 10^{-11}\right)\ by\ \ only\ \ 0.16\%,$$

confirming the high precision agreement between the geometric derivation and experimental observation.

4.5 Summary

The gravitational constant G is a derived geometric quantity originating from the bridge between the space divergence strength (Q) and the space–mass coupling (M).
It represents the macroscopic elasticity of space, equivalent to the inverse of its energy density.
Through the pair c, μ_g, the entire gravitational constant system can be reconstructed without introducing any additional empirical constants.

At this point, the gravitational layer of the natural constant hierarchy is fully established, providing the geometric foundation for the electromagnetic layer and quantum layer that follow.

5. Geometric Coupling of the Electromagnetic Layer and the Mass–Charge Bridging Constant

5.1 Geometric Origin of the Electromagnetic Constants

In vacuum, electromagnetic waves satisfy

$$c^2 = \frac{1}{\varepsilon_0 \mu_0},$$

where ε_0 and μ_0 are the vacuum permittivity and permeability, respectively.
Within the two-constant framework, c is already a defined quantity; hence ε_0 and μ_0 cannot be treated as independent constants.

They must instead be derived from the geometric flow structure of space.

In the helical space model,

- The radial component of the flow corresponds to the electric-field projection (E), and
- The tangential rotation corresponds to the magnetic-field projection (B).

The relation

$$\frac{E}{B} = c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}},$$

indicates that ε_0 and μ_0 are projection coefficients of the helical geometry of space on the electromagnetic plane.

5.2 Introduction of the Mass–Charge Bridging Constant

The spatial flow carries a mass current density characterized by μ_g.
If the temporal variation of the mass current corresponds to an electric current, then we define the mass–charge transduction constant k_{mq} as:

$$q = k_{mq} \frac{dM}{dt},$$

where q is the charge quantity.
The constant k_{mq} thus establishes a one-to-one correspondence between the rate of mass flow and the electric current.

Its dimension is: $k_{mq} = [C\, s/kg]$.

5.3 Two-Constant Expression for the Vacuum Permittivity

From the conservation of geometric–electric coupling, we obtain:

$$(E7) \quad \varepsilon_0 = \frac{1}{4\pi k_{mq}^2 \mu_g}.$$

Consequently, using $c^2 = 1/(\varepsilon_0\mu_0)$, the vacuum permeability automatically becomes:

$$\mu_0 = 4\pi k_{mq}^2 \frac{\mu_g}{c^2}.$$

Thus, both ε_0 and μ_0 are determined by μ_g, k_{mq}, and c—

showing that the electromagnetic constants share the same geometric origin as the gravitational constant.

5.4 Geometric Form of the Fine-Structure Constant

The fine-structure constant is traditionally defined as:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}.$$

Substituting $\varepsilon_0 = (4\pi k_{mq}^2\mu_g)^{-1}$ gives:

$$(E8) \quad \alpha = \frac{(ek_{mq})^2\mu_g}{\hbar c}.$$

This shows that α depends entirely on the two fundamental constants c, μ_g, the elementary charge e, and the bridging constant k_{mq}.

Hence, the electromagnetic layer is not independent but derived geometrically from the same foundation.

5.5 Numerical Verification (Interface with the Paper "Geometric–Quantum–Electromagnetic Closure of G")

Using CODATA 2022 constants:

$$e = 1.602176634 \times 10^{-19}\ C,$$

$$\alpha = 7.2973525693 \times 10^{-3},$$

$$\hbar = 1.054571817 \times 10^{-34}\ J\,s,$$

$$c = 2.99792458 \times 10^{8}\ m/s,$$

we can invert the above equation to solve for μ_g:

$$\mu_g = \frac{\hbar c\alpha}{(ek_{mq})^2}.$$

Adopting the experimentally calibrated value

$$k_{mq} = 7.08 \times 10^{-11}\ C\ s/kg$$

(from "Geometric–Quantum–Electromagnetic Closure of the Gravitational Constant: G = 6.685 × 10^{-11}"),

we obtain:

$$\mu_g = 1.6895 \times 10^{28}\ kg/m,$$

which matches the gravitational-layer result within 0.15%.

Thus, the electromagnetic and gravitational layers are numerically unified.

Additional explanation: Regarding the paper - The gauge difference of k_{mq} values in the geometric closed loop of the mass electric conversion constant and the fine structure constant

In earlier versions (e.g., "The Geometric Closure of the Mass–Charge Transduction Constant and the Fine-Structure Constant"),

the bridging constant was written in the simplified form:

$$k_{mq}^{(\sigma=1)} = \frac{e\,G}{c^3},$$

Substitute CODATA constant to get

$$k_{mq}^{(\sigma=4\pi)} = 3.16 \times 10^{-56}\ C \cdot s/kg$$

The dimensions and units of the results are correct, but the numerical value is better than that of this paper and the paper; Geometric quantum electromagnetic closed-loop derivation of the constant of universal gravitation: in the discriminant value $G = 6.685 \times 10^{-11}$,

The value used in the system is about 4 π times.

The reasons are:

The Paper - Geometric closed-loop of mass electric conversion constant and fine structure constant at that time, the geometric normalized constant $\sigma = 4\pi$ was not introduced into the electromagnetic gravitational bridging equation,

That is, the simplified form of spherical symmetry factor $\sigma = 1$ is adopted.

In this paper and Paper "Geometric quantum electromagnetic closed-loop derivation of the constant of universal gravitation: in the discriminant value $G = 6.685 \times 10^{-11}$", the normalization constant has been fixed to $\sigma = 4\pi$ to keep the same as the geometric parent

$$Q = 4\pi c \ell_{\Omega}^{2}, \quad G = \frac{4\pi c^2}{\mu_g}$$

Consistent.

Therefore, the canonical expression of the mass electric bridge constant should be

$$\boxed{k_{mq}^{(\sigma=4\pi)} = \frac{e\,G}{4\pi c^3}.}$$

Substitute the same constant (E, G, c) to get

$$k_{mq}^{(\sigma=4\pi)} = 3.16 \times 10^{-56}\ C \cdot s/kg$$

The relationship between the two is

$$k_{mq}^{(\sigma=1)} = (4\pi)\, k_{mq}^{(\sigma=4\pi)}.$$

Therefore, the paper; The original value of the geometric closed loop of the mass electric conversion constant and the fine structure constant is 3.97×10^{-55} which is only multiplied by the geometric normalization factor $1/(4\pi)$,

The specification value 3.16×10^{-56} used in this paper is the same as that in this paper; Used in the geometric closed-loop of mass electric conversion constant and fine structure constant

$$\mu_g = 1.6895 \times 10^{28}\ kg/m,$$

$$G = 6.685 \times 10^{-11}\ \mathrm{m}^3 kg^{-1} \mathrm{s}^{-2}$$

Maintain complete consistency.

To sum up, the difference of the Paper "The geometric closed-loop of mass electric conversion constant and fine structure constant is in this paper; Geometric quantum

electromagnetic closed-loop derivation of the constant of universal gravitation: of the discriminative value $G = 6.685 \times 10^{-11}$" is not caused by calculation error, but by different values of geometric normalization constant.

In the gauge system of unified field theory, the normalized constant is fixed as $\sigma = 4\pi$, The corresponding standard bridging constant should be

$$k_{mq} = 3.16 \times 10^{-56}\ C \cdot s/kg$$

5.6 Physical Significance

1. Unified Origin

Electromagnetic constants (ε_0, μ_0, α) are not independent, but are derived from μ_g and the geometric bridge $k_{\mathrm{m}q}$.

This establishes a common geometric source for both the electromagnetic and gravitational domains.

2. Experimental Testability

Once $k_{\mathrm{m}q}$ is measured directly in laboratory conditions,

μ_g and G can be independently deduced from purely electromagnetic experiments, forming a closed loop among electric, mass, and gravitational constants.

3. Hierarchical Structure

Layer	**Primary Constant**	**Defining Relation**	**Physical Function**
Geometric Layer	c	Speed of space propagation	Unifies space and time
Coupling Layer	μ_g	Space–mass coupling constant	Unifies space and matter
Electromagnetic Layer	$k_{\mathrm{m}q}$	Mass–charge bridging constant	Unifies charge and mass
Quantum Layer	ħ	Action per radian	Unifies action and frequency

Together these four layers constitute a complete unified structure extending from geometry to electromagnetism.

5.7 Summary

Within the two-constant framework, the electromagnetic constants ε_0, μ_0, and α are derived quantities expressible solely through μ_g, c, and the bridging constant $k_{\mathrm{m}q}$.
This demonstrates that the electromagnetic and gravitational systems share a common geometric origin and are numerically interlocked, realizing the geometric–gravitational–electromagnetic closed loop.

6. Geometric Derivation of the Reduced Planck Constant ħ

6.1 From Geometric Motion to Action

In the helical spatial-flow structure, each infinitesimal flow element simultaneously exhibits radial divergence and angular rotation.
The radial motion determines the expansion rate of space ($v_r = c$), while the rotational motion defines the local angular frequency Ω.

The space–mass coupling constant μ_g expresses the mass density per unit length of space.
Therefore, a spatial element of geometric length ℓ_Ω carries a mass:

$$M = \mu_g \ell_\Omega.$$

Through the space–time unification condition, the angular frequency and the geometric length are related by:

$$\Omega = \frac{c}{\ell_\Omega}.$$

Hence, each rotation cycle of the space–mass flow can be regarded as an energy–action unit, with its action defined as the product of energy and time.

6.2 Definition of Geometric Action

The kinetic energy of one rotating element is:

$$E = Mc^2$$

Its rotation period is:

$$T = \frac{2\pi}{\Omega} = \frac{2\pi\ell_\Omega}{c}.$$

Thus, the energy–time product per radian (i.e., one $2\pi th$ of a full rotation) is:

$$\begin{aligned} S_{1\,rad} &= \frac{E\,T}{2\pi} = Mc^2\frac{1}{\Omega} \\ &= \frac{Mc^2\ell_\Omega}{c} \\ &= Mc\ell_\Omega. \end{aligned}$$

Substitute $M = \mu_g\ell_\Omega$ to get:

$$S_{1\,rad} = \mu_g\, c\, \ell_\Omega^2.$$

6.3 Geometric Definition of the Reduced Planck Constant

We therefore identify:

$$(E4) \quad \hbar = \mu_g c\ell_\Omega^2.$$

More generally, without presupposing $\Omega = c/\ell_\Omega$, it can be written as:

$$\hbar = \mu_g\Omega\ell_\Omega^2.$$

This expression reveals the geometric origin of the reduced Planck constant:

ħ is determined solely by the two fundamental constants c, μ_g and the geometric scale ℓ_Ω, without requiring any empirical or independent assumption.

6.4 Dimensional and Logical Verification

1. Dimensional Consistency

$$[\mu_g] = kg/m,\ \ [\Omega] = \mathrm{s}^{-1},\ \ [\ell_\Omega^2] = \mathrm{m}^2.$$

The product $[kg\,\mathrm{m}^2/s] = J\,s$ is completely equal to the unit of the reduced Planck constant.

2. Logical Sequence

The constant ħ emerges from the combination of energy and angular motion, as a direct result of space–mass coupling.

Only after ħ is defined can one determine quantum energy density

$$\varepsilon_q = \hbar\Omega/\ell^3$$

which in turn leads naturally to the Planck length and Planck time.

Thus, the derivation of ħ precedes all Planck-scale quantities in logical order.

6.5 Physical Interpretation

1. Geometric Origin of Quantization

The constant ħ represents the minimal action associated with a space–mass coupling unit rotating through one radian.

Quantization therefore arises geometrically, not statistically.

2. Bridge Between Geometry and Dynamics

ħ directly links the macroscopic geometry (μ_g, ℓ_Ω) to microscopic dynamics (Ω), establishing the geometric–quantum bridge of the unified field theory.

3. Closure of Energy Density

From the relation

$$\varepsilon_q = \hbar\Omega/\ell^3 = \mu_g c^2/\ell^2,$$

the equilibrium between the zero-point energy density and the geometric energy density

$$\varepsilon_{\mathrm{geo}} = \frac{1}{2}\,\pi\mu_g c^2$$

defines the critical scale $\ell_\Omega = \ell_{\mathrm{p}}$ (the Planck length).

4. Experimental Connection

Since ħ is an SI-defined constant,

measuring μ_g allows one to reverse-calculate the geometric scale:

$$\ell_\Omega = \sqrt{\frac{\hbar}{\mu_g c}}.$$

This provides an indirect metrological path to determine the geometric length of space from quantum experiments.

6.6 Comparison with Traditional Definitions

Aspect	Traditional Physics	Geometric Definition (This Work)	Key Difference
Source of ħ	Empirical constant $\hbar = \hbar_{exp}$	Derived geometrically $\hbar = \mu_g c \ell_\Omega^2$	From empirical to geometric
Physical Meaning	Action in each quantum transition	Action per radian of a helical space unit	From statistical to structural
Independence	Independent constant	Derived from c, μ_g	Reduced fundamental set
Derivation Logic	$G, \hbar, c$ coexist	$c, \mu_g \rightarrow \hbar$	No cycle definition

6.7 Summary

The reduced Planck constant ħ originates from the angular motion of the space–mass coupled flow.

At the geometric–dynamical level it can be expressed as:

$$\hbar = \mu_g \Omega \ell_\Omega^2 = \mu_g c \ell_\Omega^2.$$

It represents the action per radian of rotation of space itself — the geometric foundation of quantization.

From the two fundamental constants c, μ_g, the entire quantum layer is naturally derived, providing the dimensional and logical basis for the subsequent derivation of the Planck cluster.

7. Two-Constant Derivation of the Planck Cluster

7.1 Point of Departure

From the previous section, the reduced Planck constant was obtained geometrically as

$$\hbar = \mu_g c \ell_\Omega^2.$$

This expression depends only on the two basic constants c, μ_g and the geometric length ℓ_Ω, which represents the spatial structural scale.

When the geometric energy density equals the quantum zero-point energy density, ℓ_Ω reaches a critical value ℓ_p, known as the Planck length.

All subsequent Planck-scale quantities can then be derived directly, without assuming G or ħ as independent constants.

7.2 Derivation of the Planck Length

In the space–mass coupling system, the geometric energy density is

$$\varepsilon_{geo} = \frac{1}{2}\pi\mu_g c^2$$

The quantum zero-point energy density is

$$\varepsilon_q = \frac{\hbar\Omega}{\ell^3} = \frac{\hbar c}{\ell^4}$$

At equilibrium $\varepsilon_{geo} = \varepsilon_q$:

$$\frac{1}{2}\pi\mu_g c^2 = \frac{\hbar c}{\ell^4}$$

Solving for ℓ gives:

$$\ell = \sqrt{\frac{2\hbar}{\mu_g c}}$$

Including the spherical-symmetry normalization factor σ = 4π, we obtain the standard form:

$$(E5) \quad \ell_P = \sqrt{\frac{4\pi\,\hbar}{\mu_g c}}$$

Thus the Planck length arises entirely from c, μ_g, ħ, with no reference to G.

If (E3) $G = 4\pi c^2/\mu_g$ is used instead of μ_g,

That is to restore the traditional form $\ell_P = \sqrt{\hbar G/c^3}$.

This shows that Planck length is a natural geometric scale derived from two basic constants.

7.3 Planck Time

The Planck time is defined as the duration for light to traverse one Planck length:

$$t_P = \frac{\ell_P}{c} = \sqrt{\frac{4\pi\,\hbar}{\mu_g c^3}}$$

It represents the minimal time interval corresponding to one radian of spatial rotation at the Planck scale — the geometric–quantum time unit.

7.4 Planck Mass

From the mother equation $M = \mu_g \ell$, substituting $\ell = \ell_p$ gives:

$$M_P = \mu_g \ell_P = \sqrt{\frac{4\pi\,\mu_g \hbar}{c}}$$

It describes the coupling quality when the space length is ℓ_P, it is the extreme scale of mass in geometry quantum system.

7.5 Planck Energy and Angular Frequency

The Planck energy is

$$E_P = M_P c^2 = c^2\sqrt{\frac{4\pi\,\mu_g \hbar}{c}} = \sqrt{4\pi\,\mu_g\,\hbar\,c^3}$$

and the Planck angular frequency is

$$\omega_P = \frac{c}{\ell_P} = \sqrt{\frac{\mu_g c^3}{4\pi\,\hbar}}$$

ω_P thus denotes the maximum intrinsic oscillation frequency of the space–mass coupling system — the upper geometric–quantum limit of vibration.

7.6 Square-Root Mass Scaling and Geometric Planck Mass

To connect different hierarchical mass scales, two characteristic quantities are defined:

1. Square-root mass scale

$$M_* = \sqrt{\frac{\mu_g \hbar}{c}} = \sqrt{4\pi}\, M_P/(4\pi)^{1/2}$$

This scale lies between the conventional Planck mass and the geometric Planck mass, bridging quantum and gravitational domains.

2. Geometric Planck mass

$$M_{geo} = \mu_g \ell_P = 4\pi M_P.$$

It incorporates the full geometric normalization factor (4π) and represents the mass closure of the entire space.

7.7 Dimensional and Logical Verification

Quantity	Two-Constant Expression	Dimension	Logical Derivation
ℓ_p	$\sqrt{4\pi\hbar/(\mu_g c)}$	[m]	From energy density equilibrium
t_p	ℓ_p / c	[s]	Spatial propagation time
M_p	$\mu_g \ell_p$	[kg]	Coupled mass
E_p	$M_P c^2$	[J]	Maximum energy
ω_P	c / ℓ_p	[s^{-1}]	Limiting angular frequency

Each equation depends solely on $c, \mu_g, \hbar$, following the logical chain:

$$c, \mu_g \Rightarrow \hbar \Rightarrow \ell_P, t_P, M_P, E_P, \omega_P.$$

Therefore, the entire Planck cluster is deduced without circular definition.

7.8 Numerical Results (for $\mu_g = 1.6895 \times 10^{28} kg/m$)

Quantity	Expression	Numerical Value	Unit
ℓ_P	$\sqrt{4\pi\hbar/(\mu_g c)}$	1.619×10^{-35}	m
t_P	ℓ_P/c	5.404×10^{-44}	s
M_P	$\mu_g \ell_P$	2.179×10^{-8}	kg
E_P	$M_P c^2$	1.961×10^{9}	J
ω_P	c/ℓ_P	1.858×10^{43}	s^{-1}

Compared with the CODATA 2022 reference values, the deviation is about $(1-3) \times 10^{-3}$, showing that this geometric–quantum system is fully consistent within current experimental precision.

7.9 Physical Interpretation

1. Non-Circularity

In the traditional framework, $\ell_P = \sqrt{\hbar G/c^3}$ requires both G and ħ as inputs.

In this geometric framework, both G and ħ are derived from c, μ_g, eliminating circular dependencies.

2. Geometric–Quantum Unity

The appearance of the Planck cluster is not a numerical coincidence but a natural consequence of the spatial–mass coupling flow.

Each constant corresponds to a specific geometric–energy equilibrium within space.

3. Experimental Implications

Future high-precision quantum interference or gravitational experiments capable of measuring μ_g or fine variations in G could directly test the completeness of this two-constant closed loop.

7.10 Summary

All Planck-scale quantities — length, time, mass, energy, and frequency — can be uniquely derived from the two fundamental constants c, μ_g together with the derived ħ.

This result eliminates the circular dependencies of the traditional three-constant system and unifies the geometric, gravitational, and quantum layers within a single dimensional closure.

8. Dimensional Closure and Logical Order

8.1 Hierarchical Structure of the Constant System

The Unified Field Theory reveals that all natural constants can be classified into four hierarchical layers according to their dimensions and physical functions:

Hierarchical Layer	Principal Constant	Defining Relation	Physical Function
Geometric Layer	c	Spatial propagation speed	Unifies space and time
Coupling Layer	μ_g	Space–mass coupling constant	Unifies space and matter
Quantum Layer	ħ	Action per radian, $\hbar = \mu_g c \ell_\Omega^2$	Unifies action and frequency
Derived Layer	G, ℓ_p, t_p, M_p, E_p, α, …	Derived from the upper layers	Unifies macro and micro physics

This hierarchical organization ensures strict causality among constants:
each layer is determined by the one above it, leaving no independent parameters.

8.2 Logical Chain of Dimensional Closure

The derivation pathway among the four layers can be expressed as:

$$\{c, \mu_g\} \Rightarrow \text{[Geometric-Coupling]M}other\ formula(E1) - (E3)\hbar$$

$$\Rightarrow [\text{Energy Balance}](E4)\ \ Geometry - \ Quantum\ \{l_P, t_P, M_P, E_P\}$$

$$\Rightarrow [\text{Bridging Relations}](E7) - (E8)\ \{\varepsilon_0, \mu_0, \alpha\}.$$

Interpretation of this causal chain:

- Geometric layer → Coupling layer: defines the structure of space and mass through c and μ_g;
- Coupling layer → Quantum layer: generates ħ as the minimal geometric action;
- Quantum layer → Derived layer: yields the Planck cluster via energy equilibrium;
- Electromagnetic layer: links to experimental constants through k_{mq}, ε_0, μ_0, and α.

This chain demonstrates that the system of natural constants is dimensionally closed, causally ordered, and free of circular dependence.

8.3 Dimensional Consistency Check

Physical Quantity	Expression in $c, \mu_g, \hbar$	Dimension	Verification
G	$4\pi\ c^2/\mu_g [\mathrm{m}^3\, kg^{-1}\, \mathrm{s}^{-2}]$		Correct
$\hbar$	$\mu_g c \ell_\Omega^2 [kg\ \mathrm{m}^2/s]$		Correct
ℓ_P	$\sqrt{4\pi\hbar/(\mu_g c)}[m]$		Correct
t_P	$\ell_P/c[s]$		Correct
M_P	$\mu_{(g)}\ell_P[kg]$		Correct
α	$(e\, k_{mq})^2\mu_g/(\hbar c)$	Dimensionless	Correct

All dimensions close self-consistently within the set c, μ_g.

No external units or empirical constants are introduced, confirming the dimensional self-sufficiency of the system.

8.4 Causal Logic of the Hierarchy

1. Geometry Determines Coupling

The spatial divergence speed c and coupling density μ_g together define the gravitational constant G.

2. Coupling Determines Action

When the space–mass coupling enters rotational motion ($\Omega = c/\ell_\Omega$), the minimal geometric action ħ naturally arises.

3. Action Determines Scale

Through the energy–density balance involving ħ, the Planck cluster (ℓ_p, t_p, M_p, E_p) is derived, defining the transition scale between macro and micro realms.

4. Scale Determines Electromagnetic Constants

Via the mass–charge bridging constant k_{mq}, the same geometric–quantum scale leads to ε_0, μ_0, and α, completing the closure at the electromagnetic level.

Together these form the causal chain:

Geometry ⇒ Gravitation ⇒ Quantum ⇒ Electromagnetism.

8.5 Logical Diagram (Simplified Overview)

```
┌──────────────────────────────────┐
│ Geometric Layer: c               │
│ (Defines spacetime metric)       │
└──────────────────────────────────┘
                  │
                  ▼
┌──────────────────────────────────┐
│ Coupling Layer: μ_g              │
│ (Defines mass scale, gives G)    │
└──────────────────────────────────┘
                  │
                  ▼
┌──────────────────────────────────┐
```

```
│ Quantum Layer: ħ                    │
│ (Defines action scale, gives ℓp, tp, Mp) │
└──────────────────────────────────────┘
                  │
                  ▼
┌──────────────────────────────────────┐
│ Electromagnetic Layer: k_mq, ε0, μ0, α │
│ (Connects with measurable constants)   │
└──────────────────────────────────────┘
```

This illustrates a one-directional, non-circular logic where each layer uniquely determines the next.

8.6 Physical and Philosophical Implications

1. Extreme Simplification of the Constant System

All natural constants are derivable from two geometric constants.

The constant system transitions from an empirical assemblage to a geometric necessity.

2. Self-Closure Without Circularity

ħ, G, and ℓ_P no longer depend on mutual definitions;

every constant follows a single logical lineage with no feedback dependence.

3. Experimentally Testable Closure

Once μ_g is directly measured, all constants of physics can be theoretically verified.

Any residual deviation would reveal the local geometric variations of space itself.

8.7 Summary

The dimensional closure establishes a complete causal hierarchy of constants:

$$\{c, \mu_g\} \Rightarrow \hbar \Rightarrow \{\ell_P, t_P, M_P, E_P, \alpha\}.$$

Here:

- c defines the metric of spacetime,

- μ_g defines the mass scale of space, and
- ħ defines the action scale of quantization.

These scales interlock seamlessly, forming a unified geometric–gravitational–quantum–electromagnetic system that is fully closed and self-consistent in both dimension and logic.

9. Experimental Deviations and Physical Significance

9.1 Comparison Between Theoretical and Experimental Values

Based on the experimental calibration given in"The Geometric Closure of the Mass–Charge Transduction Constant and the Fine-Structure Constant", the space–mass coupling constant is

$$\mu_g = 1.6895 \times 10^{28}\ kg/m,$$

from which the gravitational constant is predicted as

$$G = \frac{4\pi c^2}{\mu_g} = 6.685 \times 10^{-11}\ \mathrm{m}^3 kg^{-1} \mathrm{s}^{-2}.$$

The CODATA 2022 experimental value is

$$G_{\mathrm{exp}} = 6.6743 \times 10^{-11},$$

yielding a deviation of about 0.16 %.

Using the same μ_g value, the corresponding Planck cluster is:

Quantity	Theoretical Value	CODATA 2022	Relative Δ (%)
ℓ_P	$1.619 \times 10^{-35} m$	$1.616 \times 10^{-35} m$	+0.19
t_P	$5.404 \times 10^{-44} s$	$5.391 \times 10^{-44} s$	+0.24
M_P	$2.179 \times 10^{-8} kg$	$2.176 \times 10^{-8} kg$	+0.14
E_P	$1.961 \times 10^{9} J$	$1.956 \times 10^{9} J$	+0.26

This consistent slight overestimation is not a random error, but carries clear geometric and physical meaning.

9.2 Geometric Interpretation: Minor Correction of the Spatial Symmetry Factor

In the geometric mother equations, the spherical normalization constant is

$$\sigma = 4\pi.$$

However, the real helical-space flow may exhibit a slightly contracted solid angle.
If the effective solid angle is

$$\sigma_{eff} = (1-\delta)4\pi, \quad \delta \approx 2 \times 10^{-3},$$

then μ_g would increase by ≈ 0.2 %, and G would correspondingly decrease by ≈ 0.2 %, precisely matching the experimental difference.

Hence, the deviation can be understood as a slight non-sphericity of the real spatial helix, suggesting that actual space geometry is marginally flatter or denser than the idealized model — a manifestation of higher-order geometric corrections.

9.3 Physical Interpretation: Effective Coupling of Spatial Energy Density

The gravitational constant reflects the "softness" of space, proportional to the inverse of its energy density:

$$\frac{1}{G} \propto \frac{\mu_g}{4\pi c^2}.$$

If the experimental environment has variations in temperature, magnetic field, or local mass density, the effective μ_g will be slightly larger than its ideal geometric value, resulting in a smaller measured G.

This means that laboratory measurements of G actually represent the local average coupling state of space, whereas the theoretical G denotes the global extreme-coupling limit.

This supports the UFT prediction of spatially incomplete coupling in real space.

9.4 Quantum Interpretation: Geometric Extremum vs. Observational Mean

At the quantum level, ħ is defined as a fixed constant.

If μ_g increases slightly, all Planck quantities (ℓ_p, M_p, etc.) vary as

$$\mu_g^{-}½ or \mu_g^{+}½.$$

The observed systematic positive deviation therefore implies that the actual coupling strength of space exceeds the laboratory average, suggesting that our current measurements have not yet reached the full geometric-extreme state.

This provides a quantitative target for next-generation high-precision determinations of G at the 10^{-6} level.

9.5 Implications for Precision Quantum Measurements

1. A New Baseline Constant μ_g

If μ_g is indeed ≈ 0.2 % larger than the reverse-calculated value,

it can serve as a new geometric–quantum reference constant,

linking to quantum metrology through e and $k_{\mathrm{m}q}$.

2. Goal for High-Precision G Measurements

The current experimental relative uncertainty of G is $\approx 2 \times 10^{-5}$.

Reducing this to 10^{-7} would directly test the predicted 0.2 % geometric deviation.

3. Atomic-Clock Frequency Ratios and Planck Angular Frequency

The theoretical Planck angular frequency ω_P is ≈ 0.2% higher than the standard value, this deviation can be observed in the order of 10^{-6} in the frequency comparison experiment of optical lattice clock with 10^{-18} stability.

4. Vacuum Energy and Casimir-Effect Tests

The zero-point energy density correlates with ω_P.

If ω_P is slightly higher, the predicted Casimir-force constant K_C would be larger.

Nanoscale force-measurement experiments with 10^{-4} precision could test this prediction.

5. Macroscopic Quantum Interference Experiments

Present interferometer masses ($\approx 10^{-14}$ kg $\approx 10^{-6} \times M_p$) approach the threshold.

If M_p is indeed 0.2 % larger, the critical mass for decoherence rises slightly—favoring the stability of macroscopic quantum states.

9.6 Complementarity Between Theory and Experiment

Layer Theoretical Cause of Deviation Experimental Manifestation Possible Test Direction

Geometric Layer Non-spherical solid-angle factor ($\delta \approx 2 \times 10^{-3}$)G slightly smaller Ultra-precise torsion balance, cold-atom gravity tests

Coupling Layer Local density and energy fluctuations Geographic variation of measured G Ground vs. orbital gravity comparison

Quantum Layer Spatial coupling not fully saturated Planck cluster slightly larger Atomic clocks, Casimir-force experiments

Electromagnetic Layer Tiny perturbation in mass–charge bridge Micro-differences in α, ε_0, μ_0 Quantum Hall and Josephson closed-loop tests

All these deviations fall within the reach of modern experimental precision, offering practical targets for quantum-metrology and gravitational-constant research.

9.7 Summary

The theoretical value of Planck cluster predicted by the two constant framework is about $(1-3) \times 10^{-3}$ higher than the experimental value.

This difference can be explained by the slight nonspherical symmetry or incomplete local coupling of spatial spiral geometry.

The theory gives the ideal extreme value of the natural constant, and the experiment

reflects the average state of the real space.

In the future, 10^{-6} level gravitational constant measurement, 10^{-18} level atomic clock frequency comparison and Casimir force experiment, it is expected to directly verify the accuracy limit of this geometric quantum unified model.

10. Conclusions and Outlook

10.1 Research Conclusions

1. Establishment of Two Fundamental Constants

Starting from the helical geometry of space, this paper demonstrates that all natural constants can be derived from two fundamental constants—the speed of light (c) and the space–mass coupling constant (μ_g).

The constant c defines the spacetime metric, while μ_g defines the mass scale of space. Together, they form the minimal independent generating set of physical constants.

2. Three Geometric Mother Equations (E1–E3)

$$Q = 4\pi c \ell_\Omega^2, \qquad M = \mu_g \ell_\Omega, \qquad G = \frac{4\pi c^2}{\mu_g}.$$

These equations unify the spatial divergence strength, mass coupling, and the geometric origin of the gravitational constant.

They show that G is a derived, not a fundamental, constant.

3. No-Rescaling Lemma

$$\frac{M}{\ell_\Omega} = \mu_g = \text{constant}.$$

This ensures that μ_g remains invariant across all scales, providing geometric closure to the system of natural constants.

4. Geometric Definition of the Reduced Planck Constant

From the angular motion of space,

$$\hbar = \mu_g c \ell_\Omega^2,$$

showing that ħ is the action per radian, jointly defined by μ_g and c — the geometric origin of quantization.

5. Two-Constant Derivation of the Planck Cluster

$$\ell_P = \sqrt{\frac{4\pi\hbar}{\mu_g c}},$$

$$t_P = \frac{\ell_P}{c},$$

$$M_P = \mu_g \ell_P,$$

$$E_P = M_P c^2.$$

All Planck quantities are uniquely determined by c, μ_g, ħ, without circular dependence.

6. Introduction of the Electromagnetic Layer

Through the mass–charge bridging constant $k_{\mathrm{m}q}$, the vacuum constants and fine-structure constant acquire geometric definitions:

$$\varepsilon_0 = \frac{1}{4\pi k_{mq}^2 \mu_g},$$

$$\alpha = \frac{\left(e k_{mq}\right)^2 \mu_g}{\hbar c}.$$

Thus, ε_0, μ_0, α, and G are coherently unified within the two-constant framework.

7. Experimental Consistency

Using the calibrated value

$$\mu_g = 1.6895 \times 10^{28}\ \mathrm{kg/m},$$

the predicted gravitational constant

$$G = 6.685 \times 10^{-11}\ \mathrm{m}^3 k\mathrm{g}^{-1}\mathrm{s}^{-2}$$

differs from the CODATA 2022 value (6.6743×10^{-11}) by only 0.16 %.

The Planck cluster shows a systematic 0.1–0.3 % excess, implying that the geometric calibration gives the theoretical extreme, while experimental values reflect the average state of real space.

10.2 Physical and Philosophical Significance

1. Extreme Simplification of the Constant System

The vast collection of empirical constants collapses into a system generated solely by two geometric constants, c, μ_g.

The natural-constant system thus transforms from an empirical mosaic into a geometric necessity, realizing the principle of minimality.

2. Geometric–Quantum Integration

The speed of light defines the structure of spacetime;

the space–mass coupling constant defines the structure of matter;

their interaction gives rise to the quantum action ħ.

Geometry and quantization are thereby unified, not opposed.

3. Common Origin of Electromagnetism and Gravitation

Through $k_{\mathrm{m}q}$, the electromagnetic layer and gravitational layer share a common geometric foundation.

The vacuum permittivity and permeability become projection coefficients of the spatial helix,

while the fine-structure constant α becomes the experimental indicator of the geometric closure.

4. Non-Circular and Self-Contained System

All constants follow a one-way logical chain:

$$\{c, \mu_g\} \Rightarrow \hbar \Rightarrow \{\ell_P, t_P, M_P, E_P, \alpha, G, \dots\}.$$

There is no mutual definition or feedback dependence — the system is fully closed in both dimension and logic.

5. Geometric Interpretation of Natural Constants

Each constant corresponds to a distinct feature of the spatial helix:

- c: geometric propagation rate of space;
- μ_g: linear mass density of space;
- ħ: action per radian of rotation;
- G: spatial elasticity (softness);
- α: electromagnetic projection factor.

Together, they describe the geometric kernel of the physical world.

10.3 Complementarity with the Paper "Geometric Closure of the Mass–Charge Transduction Constant and the Fine-Structure Constant"

Direction This Paper (Theoretical) "Mass–Charge Transduction Constant & Fine-Structure Constant" (Experimental)

Starting Point Derives the two-constant system geometrically Determines μ_g experimentally from electromagnetic constants

Function Provides theoretical closure and logical structure Provides experimental interface and numerical calibration

Core Result $G = 4\pi c^2/\mu_g$; complete Planck closure $\mu_g = e/(ck_{mq})$; matches experiment

Deviation Interpretation Geometric value slightly higher Experimental mean slightly lower

Unified Meaning Theoretical closure of geometry–gravity–quantum–EM

Experimental verification pathway

Together, the two papers form a dual closure system—

Paper J provides definition and logic,

Paper K provides calibration and measurement,

jointly confirming the physical and experimental validity of μ_g.

10.4 Future Research and Experimental Prospects

1. Precision Measurement of G

Target relative uncertainty $< 10^{-7}$ to verify the 0.2 % geometric deviation.

Recommended platforms: cold-atom interferometry, superconducting torsion balances, and satellite-based experiments.

2. Direct Measurement of the Mass–Charge Constant k_{mq}

Conduct high-precision mass–charge conversion experiments (e.g., comparing single-

electron mass with electric current) to determine $k_{\mathrm{m}q}$ and infer μ_g through purely electromagnetic means.

3. Quantum Frequency and Atomic-Clock Comparisons

Compare theoretical Planck angular frequency ω_P with ultra stable optical clock ratios to search for $10^{-6} - 10^{-7}$ level deviations.

4. Vacuum Energy and Casimir-Effect Tests

Employ high-sensitivity nanoscale force detectors to measure changes in zero-point energy density, testing the predicted slight increase in K_C due to higher ω_P.

5. Macroscopic Quantum Interference Experiments

Extend levitated-particle interferometry to masses of $10^{-13} - 10^{-12}$ kg, exploring the crossover region between quantum and gravitational domains.

6. Cosmological Verification

Astronomical observations may test whether μ_g varies with spatial curvature or redshift, probing whether the constancy of space–mass coupling is itself a cosmic invariant.

10.5 Epilogue

The speed of light (c) and the space–mass coupling constant (μ_g) together constitute the geometric foundation of the physical universe.
From them, all gravitational, quantum, and electromagnetic constants can be deduced, forming a self-contained and non-empirical dimensional loop.

This two-constant framework reconstructs the logical order of natural constants and provides a testable route toward both quantum gravity and ultra-precise metrology.
As experimental accuracy continues to advance, the direct measurement of μ_g may one day confirm the principle of geometric calibration, marking the final verification of the Unified Field Theory's geometric foundation.

Appendix A. Main Symbols and Definitions

Symbol	Definition / Meaning	Dimension
c	Speed of light; propagation velocity of space (spacetime metric constant)	$m \cdot s^{-1}$
μ_g	Space–mass coupling constant; mass density per unit length of space	$kg \cdot m^{-1}$
ℓ_Ω	Geometric scaling length; characteristic radius of the helical spatial flow	m
Q	Spatial divergence strength (volumetric flow rate)	$m^3 \cdot s^{-1}$
M	Coupled mass (mass equivalent of spatial length ℓ_Ω)	kg
G	Gravitational constant (geometrically derived quantity)	$m^3 \cdot kg^{-1} \cdot s^{-2}$
ħ	Reduced Planck constant (action per radian)	J·s
ℓ_p	Planck length (geometric–quantum equilibrium scale)	m
t_p	Planck time	s
M_p	Planck mass	kg
E_p	Planck energy	J
ω_P	Planck angular frequency	s^{-1}
k_{mq}	Mass–charge bridging (transduction) constant	$C \cdot s \cdot kg^{-1}$
ε_0, μ_0	Vacuum dielectric constant and permeability	$F \cdot m^{-1}, H \cdot m^{-1}$
α	Fine-structure constant	—

Appendix B. Index of Core Equations

Equation No.	Content Relation / Definition	
(E1)	$Q = 4\pi c \ell_\Omega^2$	Spatial divergence strength
(E2)	$M = \mu_g \ell_\Omega$	Space–mass coupling
(E3)	$G = \frac{4\pi c^2}{\mu_g}$	Geometric origin of gravitational constant
(E4)	$\hbar = \mu_g c \ell_\Omega^2$	Definition of reduced Planck constant
(E5)	$\ell_P = \sqrt{\frac{4\pi\hbar}{\mu_g c}}$	Planck length
(E6)	$t_P = \frac{\ell_P}{c}, \quad M_P = \mu_g \ell_P$	Derived Planck quantities

(E7) $\varepsilon_0 = \frac{1}{4\pi k_{mq}^2 \mu_g}$ Vacuum permittivity (geometric form)

(E8) $\alpha = \frac{(ek_{mq})^2 \mu_g}{\hbar c}$ Fine-structure constant (geometric form)

Standard Calibrated Constants

(adopted from "Geometric Closure of the Mass–Charge Transduction Constant and the Fine-Structure Constant"):

$$\mu_g = 1.6895 \times 10^{28}\ \mathrm{kg/m},$$

$$G = 6.685 \times 10^{-11}\ \mathrm{m^3 kg^{-1} s^{-2}}.$$

Derived Planck Quantities

$$\ell_P = 1.619 \times 10^{-35}\ \mathrm{m},$$

$$t_P = 5.404 \times 10^{-44}\ \mathrm{s},$$

$$M_P = 2.179 \times 10^{-8}\ \mathrm{kg}.$$

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